Circuit Lab, Division B

Practice Test 5 (State/National level) · written test · 50 minutes · 50 questions

Team name: _____ Team number: _____

Names: _____

Instructions

- Write all answers on the answer sheet at the back. Only the answer sheet is graded.
- Give numeric answers in metric (SI) units with the correct number of significant figures.
- You may use two calculators and your own paper notes.
- Use $k = 8.99 \times 10^{9}\ \text{N·m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19}\ \text{C}$. Wires and meters are ideal unless the question says otherwise.
- In logic questions, A′ means NOT A, AB means A AND B, and A + B means A OR B.

Section	Questions
Section I: Multiple choice	1–15
Section II: Short answer and calculations	16–37
Section III: Circuit problems	38–50

Section I: Multiple choice

Circle the one best answer. 1 point each.

1. Besides his circuit laws, Kirchhoff worked with Robert Bunsen in Heidelberg to found:
   - **A.** The theory of the electron
   - **B.** The first electric telegraph
   - **C.** Boolean algebra
   - **D.** Spectroscopy: naming elements from the light they give off

2. Before he built the pile, Alessandro Volta discovered which gas, bubbling up from marshes (1776-1778)?
   - **A.** Oxygen
   - **B.** Hydrogen
   - **C.** Neon
   - **D.** Methane

3. Charles-Augustin de Coulomb trained for, and first worked in, which career?
   - **A.** Medical doctor
   - **B.** Priest
   - **C.** Telegraph operator
   - **D.** Military engineer

4. George Boole was largely self-taught and had no university degree. In 1849 he became the first professor of mathematics at Queen's College in:
   - **A.** Lincoln
   - **B.** Cambridge
   - **C.** Cork
   - **D.** Paris

5. Two neutral metal spheres A and B touch each other. A positive rod is held near A (it never touches). With the rod still in place, B is moved away from A. Then the rod is taken away. Now:
   - **A.** A is positive and B is negative
   - **B.** Both are neutral
   - **C.** A is negative and B is positive
   - **D.** Both are positive

6. Extra charge sits on a pear-shaped metal conductor. Where is the charge per area (and the field just outside) the greatest?
   - **A.** At the flattest, widest part
   - **B.** At the sharpest (most pointed) end
   - **C.** Spread evenly over the surface
   - **D.** In the center of the metal

7. Power companies send power over long lines at very high voltage mainly because:
   - **A.** High voltage travels faster than low voltage
   - **B.** High voltage is safer for workers
   - **C.** Wires have less resistance at high voltage
   - **D.** For the same power the current is smaller, so the I^2R loss in the lines is much smaller

8. A US electric clothes dryer needs 240 V. Inside the home's panel it is connected to:
   - **A.** The two hot wires (each is 120 V from neutral, on opposite sides)
   - **B.** One hot wire and the ground wire
   - **C.** One hot wire and the neutral wire
   - **D.** Three hot wires of a 3-phase supply

9. Two copper wires have the SAME MASS. Wire Y is twice as long as wire X. What is R_Y/R_X?

A. 1/2 **B.** 2

C. 4 **D.** 8

10. A microcontroller drives a 5 V relay coil, and the relay contacts switch a 120 V lamp. What is the main reason to use the relay?

A. The relay makes the lamp brighter

B. The relay changes the 120 V AC into 5 V DC

C. A small, low-voltage circuit switches a separate high-power circuit, and the two are electrically isolated

D. The relay limits the lamp current to 5 mA

11. A stair light is switched from three places: 3-way switch A, 4-way switch B and 3-way switch C. The light is OFF when A = B = C = 0, and flipping any ONE switch always changes the light. Which expression gives the light?

A. ABC **B.** A + B + C

C. A ⊕ B ⊕ C **D.** (A ⊕ B)C

12. An old house has 2-wire outlets with no ground wire. Why can a GFCI outlet still protect people there?

A. It makes its own ground wire inside

B. It compares the hot and neutral currents, which does not need a ground wire

C. It lowers the voltage to 12 V

D. It cannot; a GFCI needs a ground wire to work

13. On the diode-test setting, a meter shows 0.58 V with the red lead on the anode, and OL with the leads swapped. The diode is:

A. Good **B.** Shorted

C. Open **D.** Connected backwards inside

14. When a silicon PN junction is reverse biased (below breakdown), the depletion region:

A. Gets wider, and only a tiny leakage current flows

B. Gets narrower, and a large current flows

C. Disappears

D. Stays the same, and the current is 0.7 A

15. What is the smallest number of 2-input NAND gates that can make a 2-input XOR gate?

A. 3 **B.** 4

C. 5 **D.** 6

Section II: Short answer and calculations

Write the answer with units and the correct number of significant figures. 2 points each.

16. q_1 = +2.0 µC is at (0, 0) and q_2 = +2.0 µC is at (0.40 m, 0). q_3 = –3.0 µC is at (0.20 m, 0.30 m). Find the size and direction of the net force on q_3.

17. A +5.0 nC charge is at x = 0 and a –2.0 nC charge is at x = 0.20 m. (a) Find the electric potential at the midpoint (x = 0.10 m). (b) How much work is needed to bring a +1.0 nC charge from very far away to that point?

18. Three identical small metal spheres: A has +9.0 μC, B and C are neutral. A touches B, and they are separated. Then B touches C, and they are separated. Find the force between A and C when they are 0.30 m apart.

19. A battery gives 1.200 A into a 5.00 Ω load and 0.750 A into an 8.50 Ω load. Find its internal resistance r.

20. For the same battery, find its EMF.

21. You must build a pack of at least 22.0 V and at least 10.0 Ah from 3.70 V, 2500 mAh cells. Find the fewest cells in series and strings in parallel, the total number of cells, and the pack energy in Wh.

22. A 1320 W heater runs on 120 V rms AC. Find the peak current in its cord.

23. A 75 W, 120 V bulb has a cold filament resistance of 15 Ω. Find the current at the instant it is switched on. How many times larger is it than the normal running current?

24. A 30-day electric bill is $112.20 at $0.165 per kWh. Find the average power the home used.

25. (a) Give the 5 color bands for a 2.49 kΩ, ±1% resistor. (b) A 4-digit SMD resistor is marked 4991. What is its value?

26. An ideal 6.00 V battery drives a 10.0 Ω resistor. An ammeter with an internal resistance of 2.00 Ω is put in series to measure the current. What does it read, and by what percent is it below the true current?

27. Six 12.0 Ω resistors form the rim of a hexagon (a wheel). Six more 12.0 Ω resistors are spokes from each corner to the center. Find the resistance between the center and one corner.

28. A 15.0 V ideal battery drives 470 Ω in series with a parallel pair: 1.50 kΩ and an unknown R. The voltage across the parallel pair is 6.00 V. Find R.

29. A 20 A, 120 V kitchen circuit already runs a 1150 W microwave. How many 60 W lamps can be added before the breaker's rating is passed? (Ignore the 80% continuous-load rule here.)

30. A zener regulator: a 12.00 V supply, a 220 Ω series resistor, a 5.10 V zener diode across the load, and a 1.00 kΩ load. Find the current in the zener.

31. Use a K-map to find the simplest sum of products for F(A, B, C, D) = Σm(0, 1, 2, 5, 8, 9, 10).

32. Add the binary numbers 11011011 + 10110111. Give the sum in binary and in decimal.

33. A 4-bit ripple-carry adder adds A = 1011 and B = 0110 with a carry-in of 1. Give the 4-bit sum S and the carry-out.

34. F(A, B, C) = Σm(1, 2, 6, 7) is built with a 4-to-1 multiplexer. A and B are the select inputs (A is the high bit). What must each data input I_0 to I_3 be (0, 1, C or C′)?

35. In a hands-on task, the battery terminals read 8.64 V while the circuit runs. A 47.0 Ω resistor in series with the battery has 0.282 V across it. Find the power supplied by the battery.

36. Equal-brightness task: on a 5.00 V supply, a red LED (V_F = 1.90 V) has its own 150 Ω resistor. A green LED (V_F = 2.90 V) is in a separate parallel branch. What resistor must the green LED have so that both LEDs carry the same current?

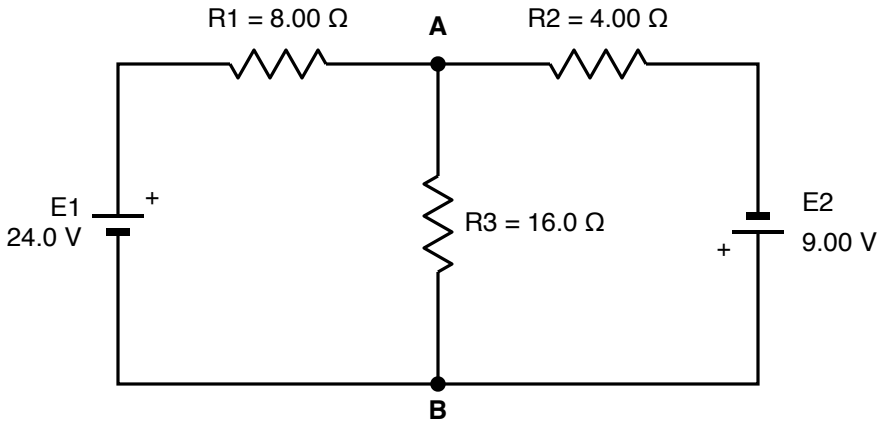
37. You have one each of 68 Ω, 150 Ω, 220 Ω and 470 Ω. Using any of them, make a resistance within 1% of 200 Ω. Show the arrangement and its exact value.

Section III: Circuit problems

Show your work. 2 points per part.

Problem A: two batteries, one reversed (Kirchhoff's laws) (Questions 38–41)

E1 = 24.0 V has its + plate at the TOP. E2 = 9.00 V has its + plate at the BOTTOM. R1 = 8.00 Ω, R2 = 4.00 Ω, R3 = 16.0 Ω. Let I_1 be the current in R1 flowing toward node A and I_2 the current in R2 flowing toward node A.



38. Write the two loop equations and find I_1.

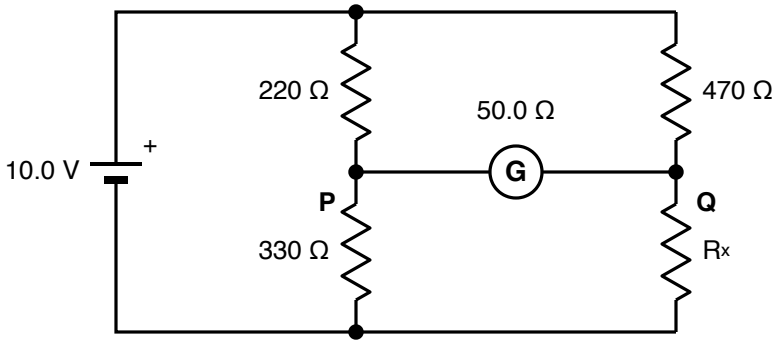
39. Find I_2. Which way does the current really flow in R2?

40. Find the current in R3 and the voltage of A relative to B.

41. How much power does E2 deliver to the circuit? (Both batteries push current the same way around the outside loop.)

Problem B: Wheatstone bridge with a galvanometer (Questions 42–45)

A 10.0 V ideal battery feeds the bridge below. P is between 220 Ω (top) and 330 Ω (bottom). Q is between 470 Ω (top) and R_x (bottom). The galvanometer G (50.0 Ω) joins P and Q.



42. What value of R_x makes the bridge balanced (no current in G)?

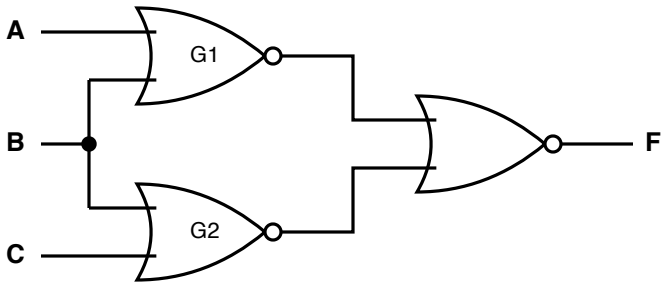
43. Now R_x = 560 Ω and G is disconnected. Find $V_P - V_Q$.

44. With R_x = 560 Ω, G is connected again. Find the current in G and its direction.

45. With G connected, find the total current from the battery.

Problem C: NOR-only logic (Questions 46–48)

Three NOR gates: G1 = NOR(A, B), G2 = NOR(B, C), and F = NOR(G1, G2).



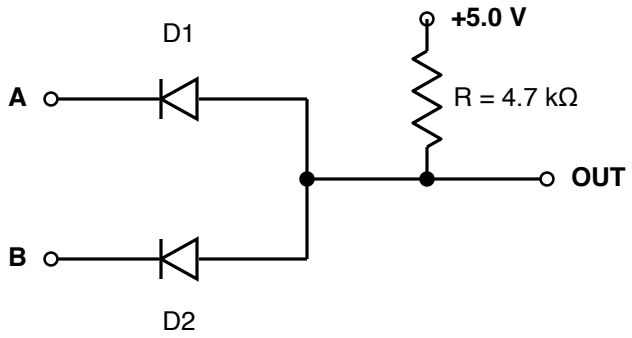
46. Write F and simplify it to the shortest sum of products.

47. Give F as a list of minterms, Σm( ).

48. Give F as a list of maxterms, ΠM( ), and write F as a product of sums.

Problem D: diode logic (Questions 49–50)

Inputs A and B are each 0 V or 5.0 V. D1 and D2 are silicon diodes (0.7 V when on), with their anodes at OUT. R = 4.7 kΩ goes from +5.0 V to OUT. Nothing else is connected to OUT.



49. Which logic gate does this circuit make (positive logic: 5 V = 1)? Explain in one line.

50. A = 0 V and B = 5.0 V. Find V_{OUT} and the current in R.

End of the test. Check your answer sheet.

Answer sheet · Practice Test 5 (State/National level)

Team name: _____ Team number: _____

Multiple choice: circle one letter. Other questions: write the answer with units.

1. A B C D

2. A B C D

3. A B C D

4. A B C D

5. A B C D

6. A B C D

7. A B C D

8. A B C D

9. A B C D

10. A B C D

11. A B C D

12. A B C D

13. A B C D

14. A B C D

15. A B C D

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